

Road Accident Risk Analysis

1. Project Overview

This project focuses on analyzing road accident data to identify high-risk locations, understand accident patterns, and evaluate key risk factors such as time, weather, traffic conditions, and vehicle involvement. The objective is to generate actionable insights to improve road safety and support data-driven decision-making.

2. Dataset Summary

Rows: 20000

Columns: 24

Key Features:

- Location: State, City, Latitude, Longitude
- Time: Date, Hour, Day of Week, Weekend, Peak Hour
- Environment: Weather, Visibility, Temperature
- Road: Road Type, Traffic Density, Lanes, Traffic Signal
- Accident: Cause, Severity, Vehicles Involved, Casualties
- Special: Festival
- Calculated: Risk Score

3. Exploratory Data Analysis using Python

We performed data pre-processing and exploration using Python:

● Data Loading

- Imported dataset using **Pandas**

● Initial Exploration

- Used `.info()` and `.describe()` to understand structure and statistics

● Data Cleaning

- Handled missing values
- Removed inconsistencies
- Standardized column names

- **Feature Engineering**

- Created:
 - Weekend Status
 - Peak hour status from is_peak_hour
 - Risk-based insights

- **Data Validation**

- Ensured data consistency across:
 - Location
 - Time
 - Risk factors

- **Database Integration**

- Loaded cleaned data into **PostgreSQL** for SQL-based analysis

4. Data Analysis using SQL (Business Problems)

We performed structured SQL analysis to answer key risk-related business questions:

1. Accident Hotspots

Identified the most accident-prone city in each state.

Data Output Messages Notifications				
	State text	City text	Total Accidents bigint	Rank bigint
1	Delhi	Delhi	2433	1
2	Karnataka	Bangalore	2438	1
3	Maharasht...	Pune	2517	1
4	Punjab	Chandigarh	2577	1
5	Tamil Nadu	Chennai	2575	1
6	Telangana	Hyderabad	2400	1

2. State-wise Risk Analysis

Calculated average risk score across states to identify high-risk regions.

Data Output Messages Notifications			
	State text	Average Risk Score numeric	Total Accidents bigint
1	Punjab	0.442	2577
2	Karnataka	0.440	2438
3	Telangana	0.439	2409
4	Tamil Nadu	0.438	2575
5	Delhi	0.437	2433
6	Maharashtra	0.435	5000

3. Accident Causes

Analysed major causes contributing to accidents.

Data Output Messages Notifications		
	Accident Cause text	Total Accidents bigint
1	Distraction	4026
2	Overspeeding	4025
3	Weather	3997
4	Drunk Driving	3978
5	Poor Road	3974

4. High-Risk Cities

Identified top cities with highest average accident risk.

Data Output Messages Notifications		
		
		
		
	City text	Average Risk Score numeric
1	Chandigarh	0.442
2	Bangalore	0.440
3	Hyderabad	0.439
4	Chennai	0.438
5	Delhi	0.437

5. Vehicle-Based Risk

Evaluated how accident risk varies with number of vehicles involved.

Data Output Messages Notifications		
		
		
		
	Vehicle Type text	Average Risk Score numeric
1	Four Wheeler	0.445
2	One Wheeler	0.438
3	Two Wheeler	0.437
4	Five or More	0.435
5	Three Wheeler	0.434

6. Weekend Urban Accidents

Measured accident frequency in urban areas during weekends.

Data Output Messages Notifi	
	
	Weekend Urban Accidents bigint
1	1946

7. Peak Hour Analysis

Analysed accidents during peak traffic hours in urban area.

Data Output		Messages	Notificat
			      
	Peak Hour Urban Accidents bigint		
1	1717		

8. Fatal Accident Causes

Identified causes leading to fatal accidents.

Data Output		Messages	Notifications
			       
	Cause text	Fatal Accidents bigint	
1	Poor Road	623	
2	Drunk Driving	618	
3	Overspeedi...	603	
4	Weather	601	
5	Distraction	542	

9. Festival vs Normal Days

Compared accident risk during festival and non-festival days.

Data Output		Messages	Notifications
			       
	Event Type text	Total Accidents bigint	
1	Normal Days	19885	
2	Festival Days	115	

10. Day-wise Risk

Determined the most dangerous day of the week.

Data Output

Messages

Notifications

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	Day text		Total Accidents bigint	
1	Monday		2966	
2	Friday		2879	
3	Tuesday		2871	
4	Saturday		2867	
5	Sunday		2856	
6	Thursday		2835	
7	Wednesday		2726	

5. Dashboard in Power BI

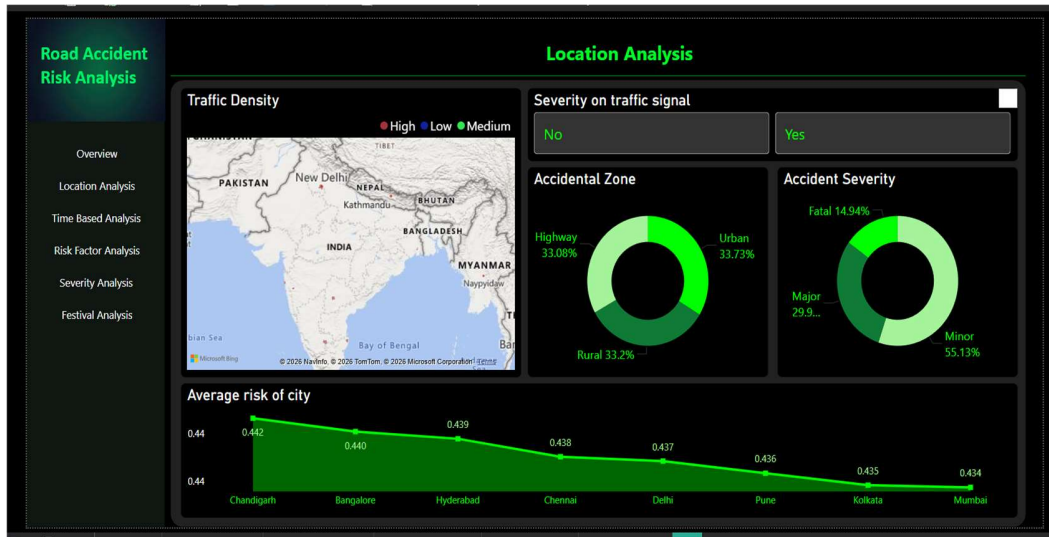
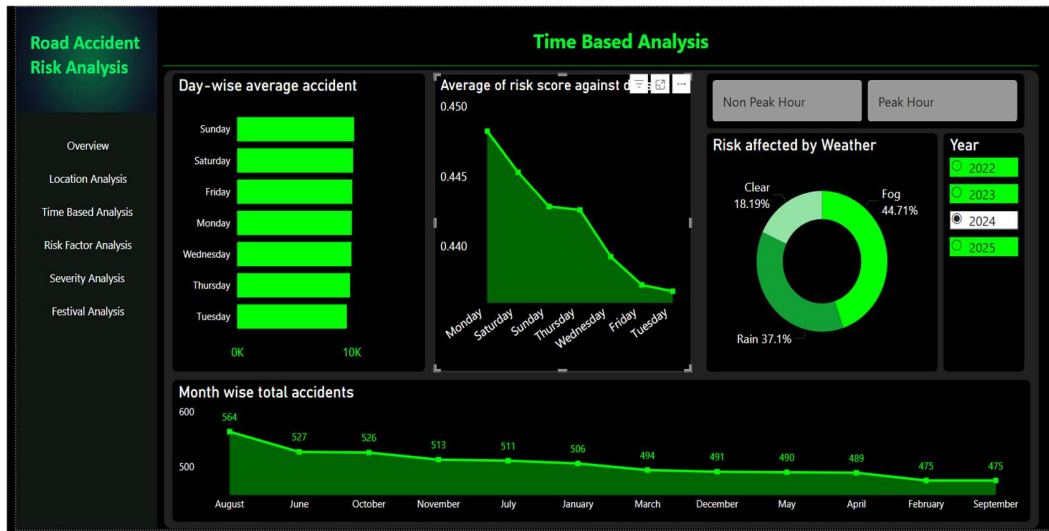
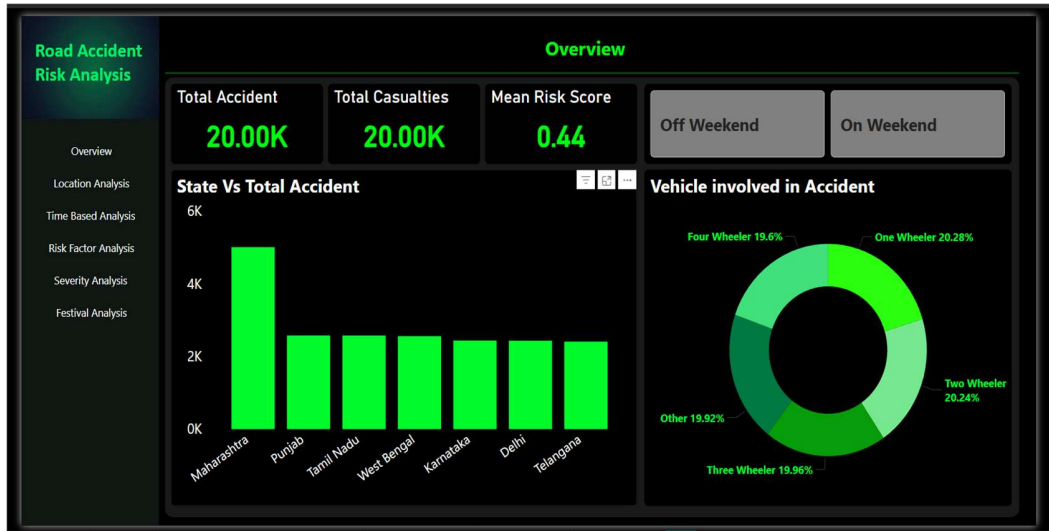
An interactive dashboard was created to visualize insights:

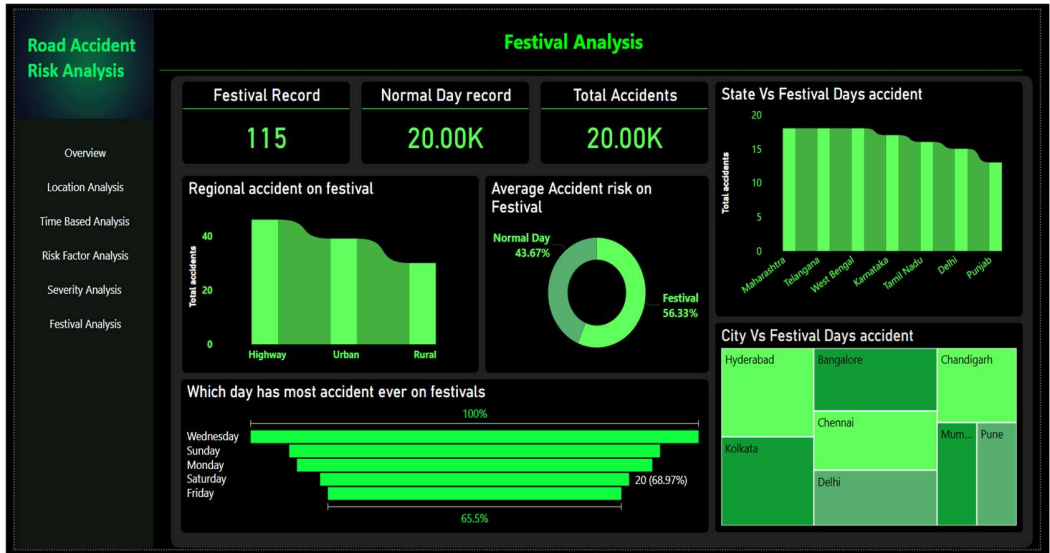
Key Visuals:

- KPI Cards (Total Accidents, Avg Risk, Fatal Cases)
- Accident Hotspot Map (Latitude & Longitude)
- State-wise Risk Comparison
- Time-based Trends (Hour, Day)
- Cause Analysis
- Vehicle Risk Analysis
- Festival vs Normal Comparison

Features:

- Dynamic filters (State, Weather, Time)
- Risk-based color coding (Red–Yellow–Green)
- Multi-page navigation





6. Business Recommendations

Improve High-Risk Areas

- Focus on accident hotspots with better infrastructure and monitoring

Traffic Management

- Implement stricter control during peak hours and weekends

Weather-Based Alerts

- Introduce warning systems during poor weather conditions

Vehicle Safety Policies

- Improve regulations for multi-vehicle traffic scenarios

Festival Traffic Planning

- Increase traffic control during festivals and public events

Awareness Campaigns

- Address major causes like over speeding and negligence